

CS 40700 Product Backlog

Product Name: Yarrow

Team Number: 13

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Problem Statement

Scanned documents often contain valuable information that is difficult to extract and search reliably because tables, figures, complex layouts, and page breaks complicate automated parsing. Existing document-processing workflows often separate document extraction, large-scale processing, cross-document search, and source verification into different steps or systems.

Our project will allow users to upload batches of scanned PDFs and images, convert their contents into structured and searchable data, and preserve traceability between extracted information and its original page and region. This will provide users with a unified workflow for processing, searching, and verifying information across large document collections.

Background Information

The target users are professionals and any individual who needs to extract, audit, and analyze tabular and visual data in large volumes of scanned reports. These users need a tool that can process the documents at scale, while allowing them to look up information across documents or locate specific text content in the original document. Although existing systems provide some of these capabilities, they do not necessarily integrate bulk document processing, cross-document search, and direct visual traceability into a single workflow.

There are existing systems that are able to parse text from documents. There are OCR web tools that are already implemented that are able to extract text data from documents and link text sections back to parts of the original documents. However, many of them do not support bulk document uploads and text search across documents. As a result, users may need to manually upload documents one at a time, which can be time consuming.

Another widely used system is multimodal LLMs. These LLM systems, such as ChatGPT, allow users to upload files as input. However, there are several limitations. First, users have to manually manage the prompt used for text extraction. For example, a user providing multiple documents needs to specify how to format the parsed results in a

single output message, then write a script manually to separate and process the results. Additionally, these LLMs may not provide the same direct traceability between extracted text and its original location, such as the source document and page from which the text was obtained.

This project would address these limitations and aim to provide a systematic way to parse documents. The proposed system would support processing of multiple documents, allowing users to bulk-upload documents to the system and process them. After a document is parsed, bounding boxes would be drawn on the original document that correspond to specific extracted texts, offering traceability and explainability. Additionally, the system would provide search capabilities that allow users to look up specific text data across documents, enabling them to efficiently look up desired information across a large volume of documents.

Requirements

Functional Requirements

Task #	Functional Requirement	Estimated Hours on Task
1	As a user, I would like to create an account	2
2	As a user, I would like to login to the account I have created	1
3	As a user, I would like to view all the documents associated with my account	3
4	As a user, I would like to edit my login credentials	1
5	As a user, I would like to delete documents associated with my account	1
6	As a user, I would like to delete my account	1
7	As a user, I would like to upload a PDF document so that its contents can be processed.	2
8	As a user, I would like to upload multiple documents at once so that I do not need to submit each file individually.	2
9	As a user, I would like to process the documents I have uploaded by parsing their	4

	text content.	
10	As a user, I want temporary worker or processing failures to not result in loss of my uploaded document or previously completed processing data	5
11	As a user, I would like to know when document processing has failed.	2
12	As a user, I want to view processed documents individually and their extracted text content	3
13	As a user, I want to view the bounding boxes on each processed document that links to certain extracted text content.	2
14	As a user, I would like to look up specific pages of processed documents by text search.	4
15	As a user, I would like to perform advanced searches across documents using filters like date ranges or document type.	3
16	As a user, I would like to export extracted content for external use, such as markdown	3
17	As a user, I would like to connect my cloud storage (e.g., Google Drive/Dropbox) to automatically import documents.	5
18	As a user, I would like to share processed documents and their bounding-box data with other users for collaborative review.	8
19	As a user, I would like to securely log out of my account.	2
20	As a user, I would like the system to preserve the original page number associated with each piece of extracted information	4
21	As a user, I would like extracted document information to be represented in structured	6

	JSON so that it can be used programmatically	
22	As a user, I would like the system to identify different document regions such as headers, paragraphs, figures, and tables.	8
23	As a user, I would like tables to be extracted while preserving their rows, columns, and structure.	8
24	As a user, I would like tables that continue across page breaks to be reconstructed as one logical table.	10
25	As a user, I would like to view the original document and its extracted structured data side-by-side.	6
26	As a user, I would like to click extracted data and have the corresponding location highlighted on the original document.	6
27	As a user, I would like search results to show the document, page, and matching extracted content where the result originated.	5
28	As a user, I would like to click a search result and navigate directly to the corresponding location in the source document.	5
29	As a user, I would like failed document-processing jobs to be retried without uploading the document again.	5
30	As a user, I would like to view the progress of a document-processing job while it is running.	4
31	As a user, I would like to view the overall progress of a batch containing multiple documents.	5
32	As a user, I would like to receive warnings that my document may be blurred instead	3

	of silently receiving poor extraction results.	
33	As a user, I would like to receive warnings that my document may be skewed instead of silently receiving poor extraction results.	3
34	As a user, I would like to receive warnings that my document may be cropped/truncated instead of silently receiving poor extraction results.	3
35	As a user, I would like to receive warnings that my document may be too dark/too light instead of silently receiving poor extraction results.	3
36	As a user, I would like to upload PNG and JPEG document scans in addition to PDFs.	5
37	As a user, I would like unsupported file types to be rejected with an explanation.	3
38	As a user, I would like to view document metadata including file name, page count, upload date, and processing status.	4
39	As a user, I would like to rename uploaded documents.	3
40	As a user, I would like to sort my documents by name, upload date, or status.	3
41	As a user, I would like to filter my document library by processing status.	3
42	As a user, I would like to cancel a queued document-processing job.	5
43	As a user, I would like to reprocess a document without uploading it again.	4
44	As a user, I would like the system to preserve the correct reading order in multi-column documents.	10
45	As a user, I would like rotated pages to be detected and corrected before OCR.	8

46	As a user, I would like extracted text to include a confidence score.	6
47	As a user, I would like low-confidence extracted regions to be visually marked for review.	5
48	As a user, I would like figure captions to remain associated with their figures.	8
49	As a user, I would like extracted table data to be exportable as CSV.	5
50	As a user, I would like extracted data to be exportable as JSON.	4
51	As a user, I would like to manually correct incorrectly extracted text.	8
52	As a user, I would like my corrections to be saved.	5
53	As a user, I would like to toggle bounding boxes on and off.	2
54	As a user, I would like bounding boxes to remain aligned while zooming the document.	6
55	As a user, I would like search matches highlighted after navigating to their source page.	4
56	As a user, I would like search results to include a short surrounding text snippet.	4
57	As a user, I would like to perform exact-phrase searches.	4
58	As a user, I would like to search for extracted table contents.	6
59	As a user, I would like to sort search results.	3
60	As a user, I would like to receive a notification when a long-running batch completes.	4

61	As a developer, I would like automated unit tests for critical backend API functionality.	10
62	As a developer, I would like automated tests for OCR and document-processing functionality.	10
63	As a developer, I would like integration tests covering document upload, processing, storage, and retrieval.	12
64	As a developer, I would like integration tests for table and layout extraction using known test documents.	12
65	As a developer, I would like end-to-end tests covering major user workflows.	12
66	As a developer, I would like tests to run automatically when code changes are merged.	12
67	As a user, I would like to reset my password if I forget it.	5
68	As a user, I would like to search my document library by document name	4
69	As a user, I would like to view information about why a document-processing job failed so that I can determine how to correct the problem.	5
70	As an administrator, I would like to view a list of all accounts with name, email, join date, and document count	2
71	As an administrator, I would like to disable a user's account so they cannot log in, while keeping their documents intact.	2
72	As an administrator, I would like to promote a user to administrator	1
73	As an administrator, I would like a dashboard showing total users, documents, storage used, and jobs by state.	3

74	As an administrator, I would like to view processing jobs across all users, including queue position, current stage, and failure reason	4
75	As an administrator, I would like to force-cancel a stuck job and retry a failed job system-wide	3
76	As an administrator, I would like to see the status of each worker node and inference server	3
77	As a user, I would like PDFs that already contain a text layer to skip OCR and reuse the embedded text and positions.	6
78	As a user, I would like to be prompted for a password when uploading an encrypted PDF instead of the job failing silently.	3
79	As a user, I would like to be warned when I upload a document identical to one I already have.	4
80	As a user, I would like to see upload progress per file and retry a failed large upload without starting over.	5
81	As a user, I would like to upload documents by dragging and dropping files or a folder.	3
82	As a user, I would like to see how much of my storage quota I have used.	3
83	As a user, I would like to see processing status for each page, not only the document as a whole.	5
84	As a user, I would like only failed pages to be retried instead of the whole document.	6
85	As a user, I would like each result to record which model version produced it so I can reprocess when models improve.	4
86	As a user, I would like to be warned when a page contains handwriting so I know extraction quality may be lower.	5

87	As a user, I would like the system to detect document language and use the appropriate OCR model.	6
88	As a user, I would like repeated headers, footers, and page numbers to be detected and excluded from paragraphs and tables.	8
89	As a user, I would like footnotes to be linked to the text they reference.	6
90	As a user, I would like figures to be extracted as image files and included in exports.	5
91	As a user, I would like each table cell to have its own bounding box.	6
92	As a user, I would like small jobs to be prioritized so they are not stuck behind a large batch.	5
93	As a user, I would like search to tolerate common OCR errors (such as 1 vs l, 0 vs O).	6
94	As a user, I would like to search within a single document.	3
95	As a user, I would like to filter search results by confidence score.	3
96	As a user, I would like my manual corrections to be reflected in search results.	5
97	As a user, I would like to search table contents by column header.	6
98	As a user, I would like page thumbnails for navigating long documents.	4
99	As a user, I would like keyboard shortcuts in the viewer (next page, next low-confidence region, toggle boxes).	3
100	As a user, I would like to see the difference between the original extraction and my correction, and undo a correction.	6

Non-Functional Requirements

Task #	Non-Functional Requirement	Estimated Hours on Task
1	As a user, I would like to set up two-factor authentication for my account to increase security.	3
2	As an administrator, I would like to scale the document processing service by running more worker nodes and inference servers	5
3	As a user, I want the application to have an intuitive web interface supporting desktop, laptop, and mobile screens.	5
4	As a user, I want the application to be responsive. This means the application should provide appropriate loading indicators or allow me to continue using other parts of the application without interruption.	5
5	As a user, I expect uploaded documents and their extracted contents to be only accessible by authorized users	3
6	As a user, I expect my authentication credentials to be securely stored and transmitted	4
7	As a user, I want search results across processed documents to be returned within a reasonable response time under normal system load	4
8	As a user, I want long-running OCR and document-	5

	processing jobs to run in the background so that I can continue using other parts of the application	
9	As a user, I want the failure of one document processing job to not cause other documents in the batch to fail	4
10	As a user, I want failed processing jobs to be recoverable or retryable without needing to upload the document again	4
11	As an administrator, I want additional worker nodes to be added without redesigning the document processing system so that processing capacity can scale	5
12	As a user, I want clear error messages that explain when an upload, processing job, or search operation fails	4
13	As a user, I want document processing states such as queued processing, completed, and failed to be clearly distinguishable	4
14	As a developer, I want the document processing, search, storage, and interface components to be sufficiently modular so that they can be tested and modified independently	5
15	As a user, I want the application to support current versions of major desktop browsers so that I can access	4

	the system reliably across common environments.	
16	As a user, I want core application workflows to remain usable through keyboard navigation and clearly labeled interface elements.	5
17	As a user, I expect deleting a document to also remove its associated extracted data and search-indexed content.	5
18	As a developer, I want processing failures to generate logs that identify the affected document, job, and processing stage so that problems can be diagnosed efficiently.	6
19	As a developer, I want environment-specific configuration such as credentials, storage locations, and inference endpoints to be configurable rather than hard-coded.	5
20	As a user, I want previously completed document-processing results to remain available even if a worker or inference service temporarily fails.	5
21	As an administrator, I would like to set global limits such as maximum file size and maximum documents per user, so one account can't consume all storage.	3
22	As an administrator, I would like a log of administrative actions, recording who	3

	disabled or deleted what and when	
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